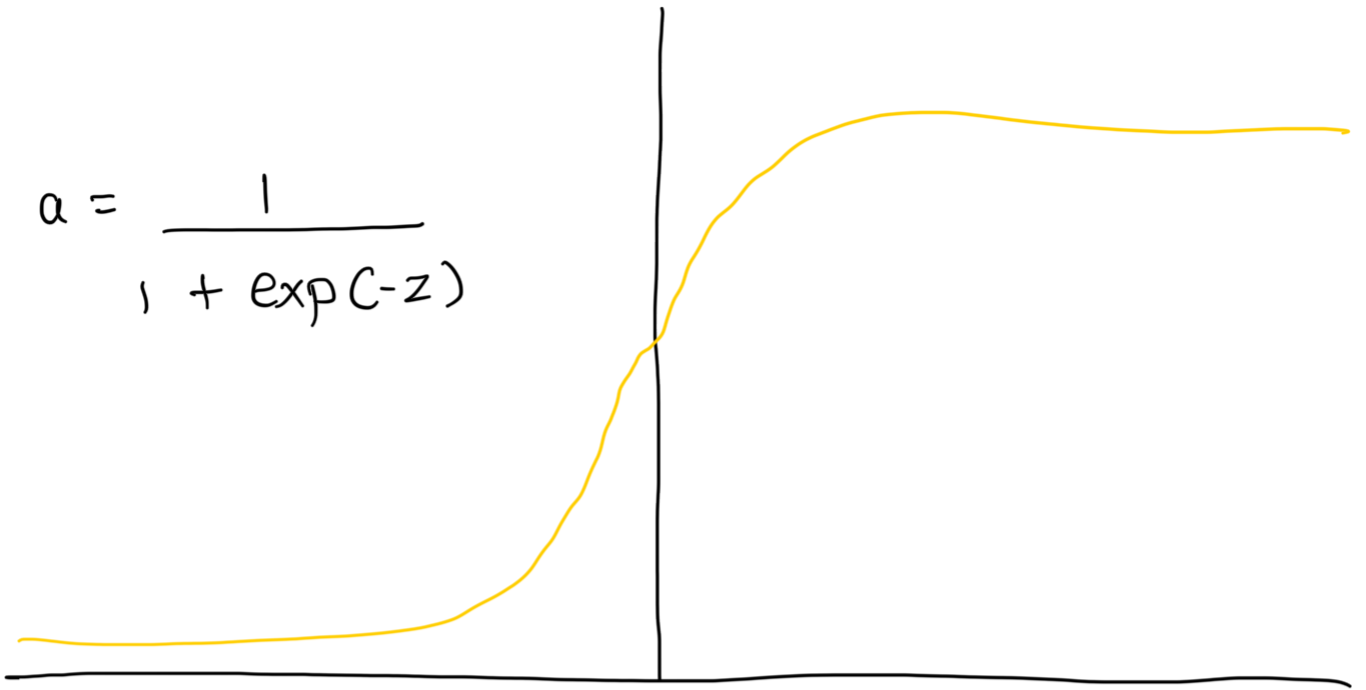


# Derivative of Sigmoid

$$a = \frac{1}{1 + \exp(-z)}$$



$$\frac{d}{dx} \left[ \frac{1}{1 + e^{-x}} \right] = \frac{d}{dx} [(1 + e^{-x})^{-1}]$$

$$= -(1 + e^{-x})^{-2} \frac{d}{dx} (1 + e^{-x})$$

$$= -\frac{e^{-x}}{(1 + e^{-x})^2} \frac{d}{dx} (-x)$$

$$= \frac{e^{-x}}{(1 + e^{-x})^2}$$

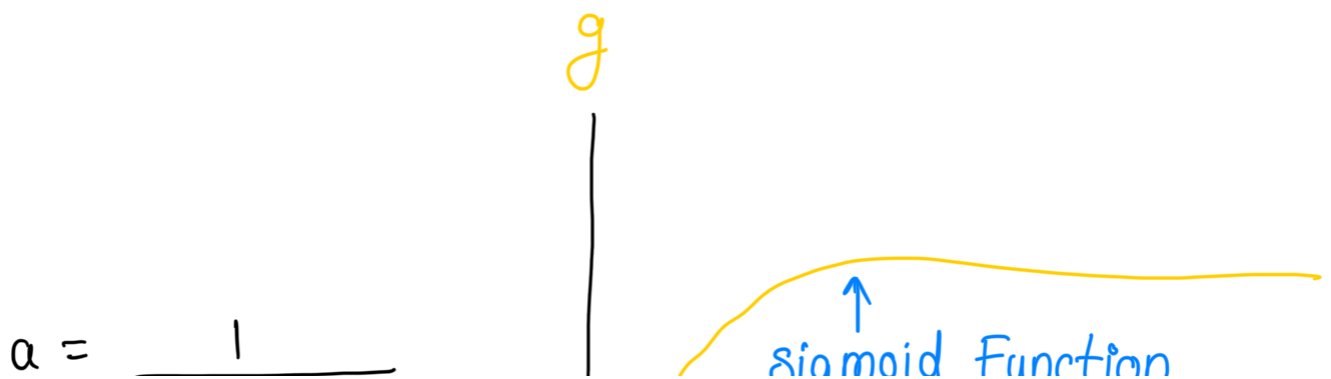
$$= \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}} \Rightarrow \sigma(x) \left[ \frac{e^{-x}}{1+e^{-x}} \right]$$

$$= \sigma(x) \left[ \frac{1+e^{-x}-1}{1+e^{-x}} \right]$$

$$= \sigma(x) \left[ \frac{\cancel{1+e^{-x}}}{\cancel{1+e^{-x}}} - \frac{1}{1+e^{-x}} \right]$$

$$= \sigma(x) [1 - \sigma(x)]$$

$$\sigma'(x) = \sigma(x) [1 - \sigma(x)]$$



$$\frac{1}{1 + \exp(-z)}$$

sigmoid function  
Graph

sigmoid Function  
derivative graph

